Dr. Jan van Roestel

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Research summary

I am an observational astronomer and focus on white dwarf binary stars, specifically systems that emit gravitational waves that will be detectable by the LISA gravitational wave satellite. I am an expert in time-domain astronomy and the use of optical time-domain survey telescopes, including the Zwicky Transient Facility and Rubin/LSST. I am also the project scientist for the BlackGEM telescope project which is aimed at finding the optical signals of Neutron-star mergers and study the variable night sky.

Current

Sept. '25 - current **Postdoc, Institute of Science and Technology Austria (ISTA)**
At ISTA, I use various large surveys to study variable white dwarfs such as binary white dwarfs that are gravitational wave sources, planets around white dwarf stars, and also massive, rapidly rotating white dwarfs that are the merger products of two white dwarfs.

May '23 - current **BlackGEM Project Scientist**
I coordinate the BlackGEM science-team and optimize the scientific output of the project.

Sept. '25 - current **Guest Researcher, University of Amsterdam (UvA)**

Previous employment and Education

July '22 - June '25 **VENI postdoctoral fellow, Astronomy and Physics, University of Amsterdam**
Working on project "The life and death of white dwarf binary stars" where I use time domain surveys, datamining methods to study double white dwarf binaries, both detached and accreting. (280 kEuro)

Sept. 2018 - May 2021 **Postdoctoral researcher, Astronomy and Physics, California Institute of Technology**
Galactic science working group lead for the Zwicky Transient Facility-collaboration; studying compact and ultracompact white dwarf binary systems; machine learning classification of variable stars.

Nov. 2013 - Aug. 2018 **PhD in Astronomy and Physics, Radboud University**
Supervisors P.J. Groot & T. Prince (Caltech) & S.R Kulkarni (Caltech)
Thesis: ***"Fast optical variability with the Palomar Transient Factory"***
For my PhD thesis, I used data from the Palomar Transient Factory (PTF) to determine the rate of intra-night transients in preparation for the search for kilonovae. In addition, I worked on data-mining PTF to find short-period white dwarf binary stars. PhD awarded 27-11-2018

Sept. 2011 - Aug. 2013 **MSc in Astronomy and Physics, *Cum Laude*, Radboud University**
supervisor P.J. Groot; Thesis topic: *A new eclipsing white dwarf - red dwarf binary star in the CV period gap.*

Sept. 2008 - Aug. 2011 **BSc in Physics, Radboud University**; BSc. thesis supervisor G. Nelemans

Collaborations and other Academic contributions

Sept '25 - current	*Cerberus* high speed camera group member (succesor of CHIMERA)
April '24 - current	Rubin Transients and Variable stars & Compact binaries team member
May '23 - current	BlackGEM project scientist
Dec. '22 - current	WEAVE-WhiteDwarf team member
March '19 - current	Zwicky Transient Facility (ZTF) collaboration; Galactic-science working group member – chair from '19-'22, representing the group within the collaboration
Sept. '18 - current	LISA Astro working group member

'23 - current	Organiser of the BlackGEM in-person team meeting twice per year & bi-weekly zoom meetings.
'19 - current	Referee for various journals (PASJ, AJ, A&A & MNRAS, etc)
'25	ESO TAC member (2x)
'23 - '25	Lunch talk co-organiser at UvA
July '21	Lead co-organiser of a two day workshop of stellar astronomy with ZTF in Warwick
May '21 - Oct. '21	Co-organising the "UV-club"; a series of 20 lectures of UV astronomy
'21 - '23	NOIRLab Time allocation committee member (4x)

Supervision

'23-'24	**MSc student**; modelling an anomalous white dwarf–red dwarf binary system
Summer '23	**ASPIRE summer student**; finding eclipsing AM CVn binaries. (paper)
Spring '23	**BSc student**; finding new quadruple eclipsing stars in ZTF (paper).
Summer '22	**ASPIRE summer student**; finding AM CVn binaries using machine learning.
Summer '21	**SURF summer student**; finding period bouncer cataclysmic variables using ZTF.
Summer '20	**SURF summer student**; finding AM CVn systems using colours of dwarf novae.
'16 -'17	**BSc. thesis co-supervisor** of two BSc. students at Radboud University

Teaching

Fall '23 & '26	**Guest lecturer**; (2hrs); MSc course "Stellar Evolution and Structure" at UvA
May '23 & '25	**Guest lecturer**; (2hrs); BSc course "Extreme Astrophysics" at UvA
April '23	**Guest lecturer** (1hr) UTFSM (Valparaiso, Chile); Intro to accreting white dwarfs for undergrads
Aug. '21	**ZTF summer school**; 15 minute talk to introduce accreting white dwarf binary stars (Youtube)
June '20	**Teaching high school students**; I designed and taught 4 weeks of lectures (2 times per week, remote).
June '19	**Teaching high school students** attending a 8 week long summerschool. I designed and taught 2 weeks of lectures (6 half-day sessions).
Aug. '14	**PTF summer school** preparing and presenting a 2hr problem session.
'13 - '16	**TA coordinator** for "Introduction in programming", a course for ∼150 first year BSc. students.
'11 - '15	**Teaching assistant** for various astronomy/physics courses (6x) at Radboud University

Science talks

March ‘26	Hotwiring2026 conference, Caltech; **invited**, *Alerts for stellar astronomy*
Jan ‘26	VLT-in-2030s conference, ESO; discussion panel member *Time domain astronomy*
June ‘25	Dutch Astronomy Conference, NL; *Planetisimal transit around white dwarfs*
November ‘24	Kupfer group, Hamburg; *Planets and planetisimal transit of white dwarfs*
July ‘24	EuroWD, Barcelona; *Cyclotron emitting white dwarf binaries*
July ‘24	COSPAR, Busan; *Cataclysmic variables found in ZTF*
July ‘24	Cosmic Transients, Stockholm; *The BlackGEM array*
May ‘24	Warsaw **seminar**; *Gravitational wave astronomy with optical survey telescopes*
March ‘24	Leuven, **seminar**; *The BlackGEM array*
Sept. ‘23	AM CVn 5 conference, Armagh; *Eclipsing AM CVn stars from ZTF*
June. ‘23	Accretion conference, Vasto; *Low accretion rate CVs*
Feb. ‘23	Armagh Observatory; **seminar**, *Exploring the population of compact binaries with ZTF*
Dec. ‘23	WD workshop, KITP; *Studying AM CVn stars with timedomain surveys*
Oct. ‘23	WD workshop, KITP; *Eclipsing white dwarf - red dwarf binaries*
Sept. ‘22	AM CVn 4.5 online conference ; *Eclipsing AM CVn stars from ZTF*
June ‘22	ML4astro, Catania ; *Classifications of variable stars with ZTF*
‘18-‘22	ZTF biannual team-meetings; *update by the variable star science working group*
May. ‘20	North Western Univ.; **seminar**, *Exploring the population of co mpact binaries with ZTF*
March. ‘20	Texas Tech; **seminar**, *Characterization of eclipsing AM CVn binaries*
Sept. ‘20	ESO Garching; *Searching for compact binaries with ZTF*
March ‘20	Radboud Nijmegen **seminar**; *Searching for eclipsing AM CVn binaries with ZTF*
June ‘20	EAS Leiden; *Machine learning classification of ZTF variable stars*
June ‘20	EAS Leiden; *Exploring the population of compact binaries with ZTF*
Sept. ‘19	CWDB meeting Yerevan; *Exploring the population of compact binaries with ZTF*
May. ‘19	DWD meeting Copenhagen; *Searching for double white dwarfs with ZTF*
Feb. ‘18	KU Leuven; **seminar** *Rapid variability with PTF*
Feb. ‘17	UCSB DISK program; *Machine learning classification of low-mass white dwarfs.*
Oct. ‘16	“Hot-wiring the transient Universe” (Villa Nova Univ., USA), *transient rates and the Galactic foreground fog with PTF*
May ‘14	PTF team meeting Stockholm; *PTF intra-night transients*
May ‘14	Dutch Astronomy Conference; *Rapid variability with PTF*

Outreach

Feb ‘25	“A movie of the night sky”, a 45 minute long lecture for amateur astronomers (last minute substitute)
Sept. ‘24	“Multistar systems”, a 2 hour lecture at the amateur astronomer club (UvA)
April ‘24	“A movie of the night sky”, a 2 hour lecture (KNVWS Gooi en Vechtstreek)
March ‘23	“Triple and multistar systems”, a 2 hour lecture (Radboud Sterrenkunde club)
‘15&‘16	Public 45-minute talk about the transient universe for the general public.
‘16	Co-organizing the observations of the transit of Mercury.
‘11 - ‘16	Regular telescope tours for prospective students and the general public.

Computer skills

Programming: python, SHELL, LaTeX, git/github, Claude.ai (agentic programming)
Data & machine learning tools: Numpy, Scipy, pandas, XGBoost, Keras, sklearn, GPU/cupy
Astrophysics: IRAF, Astropy, lcurve, development of various reduction pipelines for optical imaging and spectroscopic data.

Technical work

‘26 - current	Cerberus; Supporting the commissioning of the new camera
‘20 - current	Develop and maintain the data reduction pipeline for the CHIMERA high-speed photometry instrument at the Hale 5m
21 April ‘23	2 week long observing run at La Silla to test BlackGEM Development & testing a new observing mode with **WASP**: fast-readout with the guide-camera.
July/August ‘17	Assisting in the assembly and commissioning of the MEERlicht telescope at the Sutherland Observatory

Observational projects and awarded telescope time

Programmes on Survey telescopes

‘23 - current	**Co-lead** of the **BlackGEM Fast Synoptic Survey**
‘18 - ‘22	**PI and Co-I** of the **ZTF deepdrilling survey** (≈100hrs/yr)
‘19 - ‘22	Proposing and executing the ZTF partnership ”Galactic plane I-band” surveys (≈100hrs) and the ZTF-TESS survey.
‘16 - ‘17	**PI of Sky2Night 2&3@PTF**. 16 nights of PTF and the 10 nights of INT time for a survey to study intra-night transients with immediate spectroscopic followup.

Other awarded telescope time

‘26	**PI**, WEAVE TOO program for BlackGEM followup (8hrs)
‘26	**PI**, WEAVE program for the study of a long-period evolved binary star (14hrs)
‘25 - current	**PI**, LCO observing program of an anomalous stripped binary star (25hrs)
‘23 - current	**Co-PI** of an ESO-NTT **ULTRACAM** programme at the NTT (8 nights)
‘26 - current	**PI** CHIMERA & NGPS observations (ISTA telescope time, 0.5 nights)
‘25	**PI** of ESO-NTT **ULTRACAM** for WD-planet transit followup (3 nights)
‘21 - ‘25	**Co-I**; **HST** project on cataclysmic variable and AM CVn stars.
‘23 - ‘24	**Co-I** of two **Hipercam** programs at the 10m GTC for AM CVN and CV followup (20hrs)
‘23 - ‘24	**Co-I**; **VLT** polarimetry observations of a sample of white dwarf binaries I discovered.
‘22 - ‘24	**Co-I**; 6 nights GTC **Hipercam** proposals on eclipsing WD-RD systems I discovered.
‘22 - ‘24	**Co-I**; 6 nights ESO-NTT **ULTRACAM** proposals on eclipsing WD-RD systems.
‘22 - ‘23	**PI** of 3 *SWIFT* TOOs to observe outbursting AM CVn stars.
‘22 - ‘23	**Co-I**; ≈ 3 nights per semester **Lick** spectroscopic followup of ZTF dwarf novae
‘20 - ‘23	**PI**; ≈ 50 hrs of **SEDM robot time** to followup ZTF WD and outbursting stars.
‘19 - ‘22	**PI 20 nights** Keck & Gemini time for followup of periodic and outbursting white dwarf binaries found with ZTF.
‘19 - ‘22	**PI 30 nights** 5m Hale telescope for followup of ZTF white dwarf binary stars.
‘17 -‘19, ‘23	**PI 25 nights** at the WHT and INT for followup of PTF/ZTF discoveries

Publications and projects

My main projects and the resulting publications are listed here with a brief description. I have published 11 first author paper and two as senior author with my students. A full list of papers on which I am an author can be found of ADS (98 refereed publications at time of writing): ADS query.

PI/first author:

present **Umcovering the population of Ultracompact white dwarf binaries using survey telescopes** Ultra-compact binaries consist of atleast one white dwarf and have orbital periods of 5-65 minutes. Their orbital period evolution is governed by angular momentum loss due to gravitational wave radiation and are the most common type of detectable LISA binary. There are detached systems (double white dwarfs), and mass-transferring systems (AM CVn binaries). The questions we aim to answer are how are these systems formed; how do gravitational waves guide their evolution, and how does accretion change this? Using ZTF, and now also BlackGEM, I have been systematically searching eclipsing systems, and keep track of both both the detach double white dwarfs and accreting systems (AM CVn systems). Because they are eclipsing, the binary properties of the system can be precisely measured, which can be used to infer the formation channel. **van Roestel** *et al. (2021a, 2022); Khalil et al. (2024);* ***van Roestel*** *et al. (2025)*

present **White dwarfs with dark-companions** Short period white dwarf binaries are born from close main-sequence binaries that go through a common-envelope phase. This creates a short period binary with a white dwarf and a low mass companion. Angular momentum loss (magnetic braking, gravitational wave radiation), moves the stars close together and forms a cataclysmic variable (or similar accreting binary). Since the white dwarfs are typically many times smaller but a lot hotter. Although these systems are typically faint, eclipses are deep and easy to identify with photometric surveys. With ZTF, I have collected a sample of over 1400 such systems. Most of them are regular white dwarf-dM systems, but there are also interesting subsets: 1) the 'hidden' population of period-bounce cataclysmic variables, 2) eclipsing magnetic white dwarfs binaries, and 3) long period white dwarfs with brown dwarfs/giant planets companions. This project has spawned numerous followup efforts focussing on various aspects like magnetic fields, brown dwarfs, cyclotron emission. **van Roestel** *et al. (2021b)*, *Full sample paper in prep.*

present **Cataclysmic variables** Cataclysmic variables are Roche lobe accreting white dwarfs with low-mass companions. The evolution of these objects is governed by gravitational waves, and they can also be used as laboratories in order to understand accretion physics. The Zwicky Transient Facility is observing thousands of CVs which stand out due to their often strong variability. To understand the population of these objects better, I have made a catalogue of all outbursting cataclysmic variables that were detected in ZTF (5000+). This catalogue will be the basis to better understand the population of these objects. In addition, by searching the ZTF data, I also more than doubled the number of highly magnetic, dormant cataclysmic variables. ***van Roestel*** *et al. (2024)*, *Dwarf novae sample paper in prep.*

2023 **Eclipsing quadruple stars** Most stars can be find in multistar systems and are gravitationally bound to other stars. Quadruple stars are systems that consist of four stars. The stars in these systems can heavily affect the evolution of some or all components. Working with BSc student T. Vaessen, we developed the method of finding double eclipsing binaries quadruples that consist of two eclipsing binaries that orbit each other. We applied the method to ZTF lightcurves and increased to know sample size by more than 50%. *Vaessen and* ***van Roestel*** *(2024)*

2021 **ZTF Scope (Source ClassificatiOn ProjEct)** The Zwicky Transient Facility is producing vast amounts of data and has collected lightcurves for 2 billion objects. The first step in doing science with this data is the identify and classify the many different types of variables. With PTF, I explore the potential of machine learning classifiers and their potential to classify variable star lightcurves. With ZTF, I lead the effort to identify and classify all persistent objects. This is a huge multi-class, multi-level classification problem with different requirements on purity and completeness. We therefore came up with an alternative approach; instead of using one machine learning classifier, we are using multiple separate classifiers which classify binaries at different levels completely independently. This allows for much greater flexibility and insight. **van Roestel** *et al. (2021c); Coughlin et al. (2021)*

2019 The **Sky2Night projects** are a three week long campaigns with where we used the PTF telescope to systematically observe 400 deg^2 at a 2-hour cadence. At the same time, we used an other telescope, the WHT telescope, or INT, P60, and Hale telescopes, to observe all new transients discovered in this field, to get an unbiased sample of transients. With the data I have determined a robust observed rate for extra-Galactic and Galactic transients. In addition, we have calculated an upper-limit to the rate of fast optical transients. This work is in preparation for the systematic search for Kilonovae (the optical counterparts to NS-NS mergers detected by LIGO/Virgo). **van Roestel** *et al. (2019a), and a chapter in my PhD thesis*

2017 **EL CVn binaries in PTF**. I developed a machine learning method to discover EL CVn binaries (eclipsing binaries containing a pre-He-WD) in PTF light curve data, more than doubling the known sample size. Using the IDS@INT (17 nights), we have determined the system parameters for all systems. We show that they behave similarly to ELM white dwarfs around pulsars, and that there should be many more to be found in our Galaxy **van Roestel** *et al. (2018)*

2017 **Characterization of an eclipsing WD+RD in the period gap** The discovery and analysis of **eclipsing white dwarf - M-dwarf binary** PTF0857. I combined high cadence photometry with phase-resolved spectroscopy to determine the system parameters using Bayesian statistics. This eclipsing white dwarf - red dwarf binary has an orbital period of 2.5 hours which puts it in the CV period gap. The canonical CV evolution theory predicts that CV will cease mass-transfer, and will look like detached systems. With our knowledge of the system parameters, we concluded that PTF0857 is not likely to be such a systems, but emerged from the common envelope as we observe it today. **van Roestel** *et al. (2017)*

Publications with a significant contributions as co-author

'25 Second author on a catalogue of AM CVn binaries Green et al. (2025)

'19-current PI/Co-I of ZTF-subsurveys: **van Roestel** et al. (2019b), Kupfer et al. (2021).

'19-current Contributions to commisioning and operation of ZTF: Bellm et al. (2019); Mahabal et al. (2019).

'21 Second author on the discovery of a new type of AM-CVn outburst:

'16-'21 Contributed as an observer both with spectroscopy and high cadence photometry, and the modelling and interpretation of the data: Kupfer et al. (2017a,b); Burdge et al. (2020); Kupfer et al. (2022)

'16 The discovery of the first "white dwarf pulsar"! I obtained high cadence spectroscopy of AR Sco which shows the rapid spectral evolution of the system (∼2 minutes): Marsh et al. (2016).

Bibliography

Jan van Roestel, Leah Creter, Thomas Kupfer, Paula Szkody, Jim Fuller, Matthew J. Green, R. Michael Rich, John Sepikas, Kevin Burdge, Ilaria Caiazzo, Przemek Mróz, Thomas A. Prince, Dmitry A. Duev, Matthew J. Graham, David L. Shupe, Russ R. Laher, Ashish A. Mahabal, and Frank J. Masci. A Systematic Search for Outbursting AM CVn Systems with the Zwicky Transient Facility. AJ, 162(3):113, September 2021a. doi:10.3847/1538-3881/ac0622.

J. van Roestel, T. Kupfer, M. J. Green, T. L. S. Wong, L. Bildsten, K. Burdge, T. Prince, T. R. Marsh, P. Szkody, C. Fremling, M. J. Graham, V. S. Dhillon, S. P. Littlefair, E. C. Bellm, M. Coughlin, D. A. Duev, D. A. Goldstein, R. R. Laher, B. Rusholme, R. Riddle, R. Dekany, and S. R. Kulkarni. Discovery and characterization of five new eclipsing AM CVn systems. MNRAS, 512(4):5440–5461, June 2022. doi:10.1093/mnras/stab2421.

J. M. Khalil, **van Roestel, J.**, E. C. Bellm, J. S. Bloom, R. Dekany, A. J. Drake, M. J. Graham, S. L. Groom, S. R. Kulkarni, R. R. Laher, A. A. Mahabal, T. Prince, and R. Riddle. Four new eclipsing accreting ultracompact white dwarf binaries found with the Zwicky Transient Facility. A&A, 683:L10, March 2024. doi:10.1051/0004-6361/202349010.

J. van Roestel, K. Burdge, I. Caiazzo, T. Kupfer, P. Mróz, T. A. Prince, A. C. Rodriguez, S. Toonen, Z. Vanderbosch, E. C. Bellm, A. J. Drake, M. J. Graham, S. L. Groom, G. Helou, S. R. Kulkarni, A. A. Mahabal, R. L. Riddle, and B. Rushome. Eclipsing white dwarf from the Zwicky Transient Facility: II. Seven eclipsing double white dwarfs. *arXiv e-prints*, art. arXiv:2505.15580, May 2025. doi:10.48550/arXiv.2505.15580.

Jan van Roestel, Thomas Kupfer, Keaton J. Bell, Kevin Burdge, Przemek Mróz, Thomas A. Prince, Eric C. Bellm, Andrew Drake, Richard Dekany, Ashish A. Mahabal, Michael Porter, Reed Riddle, Kyung Min Shin, David L. Shupe, and S. R. Kulkarni. ZTFJ0038+2030: A Long-period Eclipsing White Dwarf and a Substellar Companion. ApJ, 919(2):L26, October 2021b. doi:10.3847/2041-8213/ac22b7.

J. van Roestel, A. C. Rodriguez, P. Szkody, A. J. Brown, I. Caiazzo, A. Drake, K. El-Badry, T. Prince, R. M. R. Rich, J. D. Neill, Z. Vanderbosch, E. C. Bellm, R. Dekany, F. Feinstein, M. Graham, S. L. Groom, G. Helou, S. R. Kulkarni, T. du Laz, A. Mahabal, Y. Sharma, J. Sollerman, and A. Wold. Cyclotron emitting magnetic white dwarfs in post common envelope binaries discovered with the Zwicky Transient Facility. *arXiv e-prints*, art. arXiv:2412.15153, December 2024. doi:10.48550/arXiv.2412.15153.

T. Vaessen and **J. van Roestel**. A systematic search for double eclipsing binaries in Zwicky Transient Facility data. A&A, 682:A164, February 2024. doi:10.1051/0004-6361/202348451.

Jan van Roestel, Dmitry A. Duev, Ashish A. Mahabal, Michael W. Coughlin, Przemek Mróz, Kevin Burdge, Andrew Drake, Matthew J. Graham, Lynne Hillenbrand, Eric C. Bellm, Thomas Kupfer, Alexandre Delacroix, C. Fremling, V. Zach Golkhou, David Hale, Russ R. Laher, Frank J. Masci, Reed Riddle, Philippe Rosnet, Ben Rusholme, Roger Smith, Maayane T. Soumagnac, Richard Walters, Thomas A. Prince, and S. R. Kulkarni. The ZTF Source Classification Project. I. Methods and Infrastructure. AJ, 161(6):267, June 2021c. doi:10.3847/1538-3881/abe853.

Michael W. Coughlin, Kevin Burdge, Dmitry A. Duev, Michael L. Katz, **Jan van Roestel**, Andrew Drake, Matthew J. Graham, Lynne Hillenbrand, Ashish A. Mahabal, Frank J. Masci, Przemek Mróz, Thomas A. Prince, Yuhan Yao, Eric C. Bellm, Rick Burruss, Richard Dekany, Amruta Jaodand, David L. Kaplan, Thomas Kupfer, Russ R. Laher, Reed Riddle, Mickael Rigault, Hector Rodriguez, Ben Rusholme, and Jeffry Zolkower. The ZTF Source Classification Project - II. Periodicity and variability processing metrics. MNRAS, 505(2):2954–2965, August 2021. doi:10.1093/mnras/stab1502.

J. van Roestel, P. J. Groot, T. Kupfer, K. Verbeek, S. van Velzen, M. Bours, P. Nugent, T. Prince, D. Levitan, S. Nissanke, S. R. Kulkarni, and R. R. Laher. The Palomar Transient Factory Sky2Night programme. MNRAS, 484(4):4507–4528, April 2019a. doi:10.1093/mnras/stz241.

J. van Roestel, T. Kupfer, R. Ruiz-Carmona, P. J. Groot, T. A. Prince, K. Burdge, R. Laher, D. L. Shupe, and E. Bellm. Discovery of 36 eclipsing EL CVn binaries found by the Palomar Transient Factory. MNRAS, 475:2560–2590, April 2018. doi:10.1093/mnras/stx3291.

J. van Roestel, P. J. Groot, D. Levitan, T. A. Prince, S. Bloemen, T. R. Marsh, V. S. Dhillon, D. Shupe, and R. Laher. PTF1 J085713+331843, a new post-common-envelope binary in the orbital period gap of cataclysmic variables. MNRAS, 468:3109–3122, July 2017. doi:10.1093/mnras/stx625.

Matthew J. Green, **Jan van Roestel**, and Tin Long Sunny Wong. A catalogue of ultracompact mass-transferring white dwarf binaries. *Astronomy & Astrophysics*, 700:A107, August 2025. ISSN 1432-0746. doi:10.1051/0004-6361/202554925. URL `http://dx.doi.org/10.1051/0004-6361/202554925`.

Jan van Roestel, Eric C. Bellm, Dmitry A. Duev, Christoffer Fremling, Matthew J. Graham, Frank Masci, Lin Yan, Daniel A. Goldstein, Michael Medford, Charlotte A. Ward, S. R. Kulkarni, and Thomas A. Prince. Simultaneous Observations of the Northern TESS Sectors by the Zwicky Transient Facility. *Research Notes of the American Astronomical Society*, 3(9):136, September 2019b. doi:10.3847/2515-5172/ab459c.

Thomas Kupfer, Thomas A. Prince, **Jan van Roestel**, Eric C. Bellm, Lars Bildsten, Michael W. Coughlin, Andrew J. Drake, Matthew J. Graham, Courtney Klein, Shrinivas R. Kulkarni, Frank J. Masci, Richard Walters, Igor Andreoni, Rahul Biswas, Corey Bradshaw, Dmitry A. Duev, Richard Dekany, Joseph A. Guidry, J. J. Hermes, Russ R. Laher, and Reed Riddle. Year 1 of the ZTF high-cadence Galactic plane survey: strategy, goals, and early results on new single-mode hot subdwarf B-star pulsators. MNRAS, 505(1):1254–1267, July 2021. doi:10.1093/mnras/stab1344.

E. C. Bellm, S. R. Kulkarni, M. J. Graham, R. Dekany, R. M. Smith, R. Riddle, F. J. Masci, G. Helou, T. A. Prince, S. M. Adams, C. Barbarino, T. Barlow, J. Bauer, R. Beck, J. Belicki, R. Biswas, N. Blagorodnova, D. Bodewits, B. Bolin, V. Brinnel, T. Brooke, B. Bue, M. Bulla, R. Burruss, S. B. Cenko, C.-K. Chang, A. Connolly, M. Coughlin, J. Cromer, V. Cunningham, K. De, A. Delacroix, V. Desai, D. A. Duev, G. Eadie, T. L. Farnham, M. Feeney, U. Feindt, D. Flynn, A. Franckowiak, S. Frederick, C. Fremling, A. Gal-Yam, S. Gezari, M. Giomi, D. A. Goldstein, V. Z. Golkhou, A. Goobar, S. Groom, E. Hacopians, D. Hale, J. Henning, A. Y. Q. Ho, D. Hover, J. Howell, T. Hung, D. Huppenkothen, D. Imel, W.-H. Ip, Ž. Ivezić, E. Jackson, L. Jones, M. Juric, M. M. Kasliwal, S. Kaspi, S. Kaye, M. S. P. Kelley, M. Kowalski, E. Kramer, T. Kupfer, W. Landry, R. R. Laher, C.-D. Lee, H. W. Lin, Z.-Y. Lin, R. Lunnan, M. Giomi, A. Mahabal, P. Mao, A. A. Miller, S. Monkewitz, P. Murphy, C.-C. Ngeow, J. Nordin, P. Nugent, E. Ofek, M. T. Patterson, B. Penprase, M. Porter, L. Rauch, U. Rebbapragada, D. Reiley, M. Rigault, H. Rodriguez, **J. van Roestel**, B. Rusholme, J. van Santen, S. Schulze, D. L. Shupe, L. P. Singer, M. T. Soumagnac, R. Stein, J. Surace, J. Sollerman, P. Szkody, F. Taddia, S. Terek, A. Van Sistine, S. van Velzen, W. T. Vestrand, R. Walters, C. Ward, Q.-Z. Ye, P.-C. Yu, L. Yan, and J. Zolkower. The Zwicky Transient Facility: System Overview, Performance, and First Results. PASP, 131(1):018002, January 2019. doi:10.1088/1538-3873/aaecbe.

A. Mahabal, U. Rebbapragada, R. Walters, F. J. Masci, N. Blagorodnova, **J. van Roestel**, Q.-Z. Ye, R. Biswas, K. Burdge, C.-K. Chang, D. A. Duev, V. Z. Golkhou, A. A. Miller, J. Nordin, C. Ward, S. Adams, E. C. Bellm, D. Branton, B. Bue, C. Cannella, A. Connolly, R. Dekany, U. Feindt, T. Hung, L. Fortson, S. Frederick, C. Fremling, S. Gezari, M. Graham, S. Groom, M. M. Kasliwal, S. Kulkarni, T. Kupfer, H. W. Lin, C. Lintott, R. Lunnan, J. Parejko, T. A. Prince, R. Riddle, B. Rusholme, N. Saunders, N. Sedaghat, D. L. Shupe, L. P. Singer, M. T. Soumagnac, P. Szkody,

Y. Tachibana, K. Tirumala, S. van Velzen, and D. Wright. Machine Learning for the Zwicky Transient Facility. PASP, 131(3):038002, March 2019. doi:10.1088/1538-3873/aaf3fa.

T. Kupfer, **J. van Roestel**, J. Brooks, S. Geier, T. R. Marsh, P. J. Groot, S. Bloemen, T. A. Prince, E. Bellm, U. Heber, L. Bildsten, A. A. Miller, M. J. Dyer, V. S. Dhillon, M. Green, P. Irawati, R. Laher, S. P. Littlefair, D. L. Shupe, C. C. Steidel, S. Rattansoon, and M. Pettini. PTF1 J082340.04+081936.5: A Hot Subdwarf B Star with a Low-mass White Dwarf Companion in an 87-minute Orbit. ApJ, 835:131, February 2017a. doi:10.3847/1538-4357/835/2/131.

T. Kupfer, G. Ramsay, J. van Roestel, J. Brooks, S. A. MacFarlane, R. Toma, P. J. Groot, P. A. Woudt, L. Bildsten, T. R. Marsh, M. J. Green, E. Breedt, D. Kilkenny, J. Freudenthal, S. Geier, U. Heber, S. Bagnulo, N. Blagorodnova, D. A. H. Buckley, V. S. Dhillon, S. R. Kulkarni, R. Lunnan, and T. A. Prince. The OmegaWhite Survey for Short-period Variable Stars. V. Discovery of an Ultracompact Hot Subdwarf Binary with a Compact Companion in a 44-minute Orbit. ApJ, 851: 28, December 2017b. doi:10.3847/1538-4357/aa9522.

Kevin B. Burdge, Michael W. Coughlin, Jim Fuller, David L. Kaplan, S. R. Kulkarni, Thomas R. Marsh, Eric C. Bellm, Richard G. Dekany, Dmitry A. Duev, Matthew J. Graham, Ashish A. Mahabal, Frank J. Masci, Russ R. Laher, Reed Riddle, Maayane T. Soumagnac, and Thomas A. Prince. An 8.8 Minute Orbital Period Eclipsing Detached Double White Dwarf Binary. ApJ, 905 (1):L7, December 2020. doi:10.3847/2041-8213/abca91.

Thomas Kupfer, Evan B. Bauer, **Jan van Roestel**, Eric C. Bellm, Lars Bildsten, Jim Fuller, Thomas A. Prince, Ulrich Heber, Stephan Geier, Matthew J. Green, Shrinivas R. Kulkarni, Steven Bloemen, Russ R. Laher, Ben Rusholme, and David Schneider. Discovery of a Double-detonation Thermonuclear Supernova Progenitor. ApJ, 925(2):L12, February 2022. doi:10.3847/2041-8213/ac48f1.

T. R. Marsh, B. T. Gänsicke, S. Hümmerich, F.-J. Hambsch, K. Bernhard, C. Lloyd, E. Breedt, E. R. Stanway, D. T. Steeghs, S. G. Parsons, O. Toloza, M. R. Schreiber, P. G. Jonker, **J. van Roestel**, T. Kupfer, A. F. Pala, V. S. Dhillon, L. K. Hardy, S. P. Littlefair, A. Aungwerojwit, S. Arjyotha, D. Koester, J. J. Bochinski, C. A. Haswell, P. Frank, and P. J. Wheatley. A radio-pulsing white dwarf binary star. Nature, 537:374–377, September 2016. doi:10.1038/nature18620.